INTRODUCTION TO HTML TABLES

HTML Table Tags:

1. **<table>: The <table> tag is used to define a table in HTML. All the content of the table, including rows and data cells, should be enclosed within this tag.**
2. **<tr>: The <tr> tag is used to define a table row. A table row can contain one or more table data (<td>) or table header (<th>) cells.**
3. **<td>: The <td> tag is used to define a table data cell. It holds the content inside the table. Each row (<tr>) can have multiple <td> elements to represent each data point.**

Example Code: Simple Student Record Table

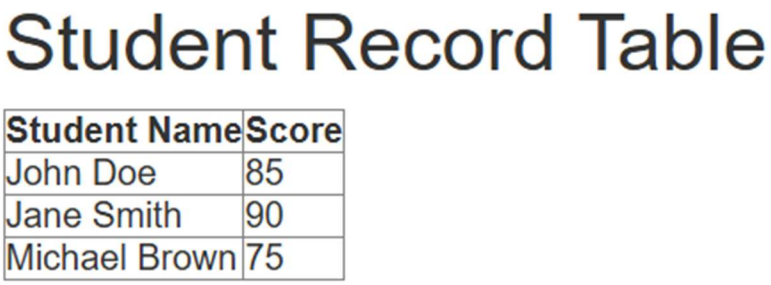
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Record Table</title>
</head>
<body>

    <h1>Student Record Table</h1>

    <table border = "1">
        <!-- Table Header -->
        <thead>
            <tr>
                <th>Student Name</th>
                <th>Score</th>
            </tr>
        </thead>
        <!-- Table Body -->
        <tbody>
            <tr>
                <td>John Doe</td>
                <td>85</td>
            </tr>
            <tr>
                <td>Jane Smith</td>
                <td>90</td>
            </tr>
            <tr>
                <td>Michael Brown</td>
                <td>75</td>
            </tr>
        </tbody>
    </table>

</body>
</html>
```

Expected Output:



EXPLANATION OF THE CODE:

1. **<table>**: This tag creates the table. All rows and data cells are contained within this tag.
2. **<thead>**: The <thead> section is used to group the header content of the table. It's optional but helps with structure. In this case, it defines the column headers: "Student Name" and "Score".
3. **<tr>**: The <tr> tag defines a table row. It is used within both the <thead> (for header rows) and <tbody> (for data rows). Each row will contain cells (<td> or <th>).
4. **<th>**: The <th> tag is used for table header cells. In this example, it defines the headers "Student Name" and "Score". Text inside <th> is usually bold by default, and it helps define what each column represents.
5. **<tbody>**: This section contains the actual data rows of the table. It's also optional but helps to separate the header from the body content, making the table easier to manage.
6. **<td>**: The <td> tag defines a table data cell. It contains the actual content for each row. For example, "John Doe" is placed in the first <td>, and "85" in the second <td>.

TABLE HEADERS WITH <TH>:

In HTML, the <th> (table header) tag is used to define a header cell in a table. It provides context or labels for the data in each column or row. When using <th>, the text is typically bold and centered by default, and it helps users understand the content of each column or row.

Example Code: Employee Directory Table

Here's a project example where we create a table listing employees, their job titles, and departments using <th> for the headers:

```
<!DOCTYPE html>
<html lang="en">
```

Employee Directory

Employee Name	Job Title	Department
John Doe	Software Developer	IT
Jane Smith	Product Manager	Marketing
Michael Brown	HR Manager	Human Resources

```
    <thead>
      <tr>
        <th>Employee Name</th>
        <th>Job Title</th>
        <th>Department</th>
      </tr>
    </thead>
    <!-- Table Body -->
    <tbody>
      <tr>
        <td>John Doe</td>
        <td>Software Developer</td>
        <td>IT</td>
      </tr>
      <tr>
        <td>Jane Smith</td>
        <td>Product Manager</td>
        <td>Marketing</td>
      </tr>
      <tr>
        <td>Michael Brown</td>
        <td>HR Manager</td>
        <td>Human Resources</td>
      </tr>
    </tbody>
  </table>

</body>
</html>
```

Expected Output:

TABLE BORDERS & STYLING USING INLINE CSS:

In HTML, we can use inline CSS to add styles directly to elements using the style attribute. To style a table, we can add borders, control the padding inside the cells, collapse borders between cells, and adjust the alignment of text.

CSS Properties:

1. **border**: This property adds a border to the table or individual cells.
2. **border-collapse**: This property is used to collapse table borders into a single border, instead of having double borders between cells.
3. **padding**: Adds space inside the cells, around the content.
4. **text-align**: Aligns the text inside the cells, typically left, center, or right.

EXAMPLE CODE: PRICE LIST TABLE WITH INLINE CSS

Here is a simple price list table with proper borders, padding, and text alignment:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Price List Table</title>
</head>
<body>

    <h1 style="text-align: center;">Price List</h1>

    <table style="width: 50%; border: 1px solid black; border-collapse: collapse; margin: 20px auto;">
        <!-- Table Header -->
        <thead>
            <tr>
                <th style="border: 1px solid black; padding: 8px; text-align: center; background-color:
#f2f2f2;">Item</th>
                <th style="border: 1px solid black; padding: 8px; text-align: center; background-color:
#f2f2f2;">Price</th>
            </tr>
        </thead>
        <!-- Table Body -->
        <tbody>
            <tr>
                <td style="border: 1px solid black; padding: 8px; text-align: left;">Apple</td>
                <td style="border: 1px solid black; padding: 8px; text-align: right;">$1.00</td>
            </tr>
            <tr>
                <td style="border: 1px solid black; padding: 8px; text-align: left;">Banana</td>
                <td style="border: 1px solid black; padding: 8px; text-align: right;">$0.50</td>
            </tr>
            <tr>
                <td style="border: 1px solid black; padding: 8px; text-align: left;">Orange</td>
                <td style="border: 1px solid black; padding: 8px; text-align: right;">$0.75</td>
            </tr>
        </tbody>
    </table>

</body>
</html>
```

Expected Output:

Price List

Item	Price
Apple	$1.00
Banana	$0.50
Orange	$0.75

Explanation of the Code:

1. **<table style="...">**:
    - width: 50%: Sets the width of the table to 50% of the parent container (the page).
    - border: 1px solid black: Adds a 1px solid black border around the table.
    - border-collapse: collapse: Collapses the table borders, so there is no space between borders of adjacent cells.
    - margin: 20px auto: Centers the table horizontally on the page with a 20px margin above and below.
2. **<th style="...">** (Table Headers):
    - border: 1px solid black: Adds a 1px solid black border around each header cell.
    - padding: 8px: Adds 8px of padding inside each header cell for spacing.
    - text-align: center: Centers the text inside each header cell.
    - background-color: #f2f2f2: Adds a light gray background color to the header cells to distinguish them from the data rows.
3. **<td style="...">** (Table Data Cells):
    - border: 1px solid black: Adds a 1px solid black border around each data cell.
    - padding: 8px: Adds 8px of padding inside each data cell for spacing.
    - text-align: left (for the "Item" column): Aligns the text to the left, which is typical for names or descriptions.
    - text-align: right (for the "Price" column): Aligns the text to the right, which is common for numerical data such as prices.

Project: Price List Table

This table displays a price list with:

- **Item names** (aligned to the left) in one column.
- **Prices** (aligned to the right) in the second column.
- Borders between each cell to visually separate the data.
- Padding inside each cell to create space around the text, making it more readable.

ROWSPAN & COLSPAN (MERGING CELLS)

Explanation of rowspan and colspan in Tables:

- **rowspan**: This attribute is used to merge multiple rows into one cell. It specifies how many rows a cell should span across.
- **colspan**: This attribute is used to merge multiple columns into one cell. It specifies how many columns a cell should span across.

These attributes are helpful when you want to organize data in a table more effectively, such as combining cells for things like "breaks" or header rows that span multiple columns.

Example Code: Class Timetable with Merged Cells for Breaks

Here's an example of a weekly class timetable where breaks are merged using rowspan and colspan to create a more readable schedule:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Class Timetable</title>
</head>
<body>

   <h1 style="text-align: center;">Weekly Class Timetable</h1>

   <table style="width: 80%; border: 1px solid black; border-collapse: collapse; margin: 20px auto;">
      <thead>
         <tr>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Time</th>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Monday</th>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Tuesday</th>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Wednesday</th>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Thursday</th>
            <th style="border: 1px solid black; padding: 8px; text-align: center;">Friday</th>
         </tr>
      </thead>
      <tbody>
         <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">8:00 AM - 9:00 AM</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Math</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Science</td>
```

```
            <td style="border: 1px solid black; padding: 8px; text-align: center;">English</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">History</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Art</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">9:00 AM - 10:00 AM</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Math</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Science</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">English</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">History</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Art</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: center;" rowspan="2">10:00 AM -
12:00 PM</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;" colspan="5">Break</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Break</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Break</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Break</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Break</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Break</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">12:00 PM - 1:00 PM</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Geography</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Math</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Physics</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Biology</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">Music</td>
        </tr>
    </tbody>
  </table>

</body>
</html>
```

Expected Output:

Weekly Class Timetable

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8:00 AM - 9:00 AM	Math	Science	English	History	Art
9:00 AM - 10:00 AM	Math	Science	English	History	Art
10:00 AM - 12:00 PM	Break				
	Break	Break	Break	Break	Break
12:00 PM - 1:00 PM	Geography	Math	Physics	Biology	Music

Explanation of the Code:

1. **Table Structure**:
    - **<table>**: The main tag to create the table.
    - **<thead>**: Contains the header row with column labels like "Time", "Monday", "Tuesday", etc.
    - **<tbody>**: Contains the data rows, where the timetable is listed.
2. **Using rowspan**:
    - In the row for the 10:00 AM - 12:00 PM time slot, the **"Time"** cell is set to span two rows using rowspan="2". This means that the "10:00 AM - 12:00 PM" time block will cover two rows, which is ideal for merging the "Time" cell across the break period.
3. **Using colspan**:
    - The "Break" cell uses colspan="5", meaning that the break spans all five days of the week (Monday to Friday). This allows the break to take up the space of five columns in a single row.
4. **Data Cells** (<td>):
    - Each cell in the timetable holds a specific class, time, or break period, aligned and bordered for clarity.

Summary:

- **rowspan** and **colspan** are used to merge cells across rows or columns.
- **rowspan** is applied to the "Time" cell to cover the break period.
- **colspan** is applied to the "Break" cell to make it span across all the days.

ADDING CAPTIONS & SUMMARIES

Explanation of <caption> and <summary> in Tables:

- **<caption>**: This tag is used to add a title or description to the table. It helps provide context to the table's content. The caption is typically placed at the top of the table and is centered by default.
- **<summary>**: This tag is used for accessibility purposes and can be placed inside a <details> element. It provides a summary or description for screen readers to understand the content of the table, especially for users with disabilities.

While <summary> is not directly related to a <table>, it can be included within a <details> element that wraps a table or other content to provide a description or summary for assistive technologies.

EXAMPLE CODE: PRODUCT COMPARISON TABLE WITH CAPTION AND SUMMARY

Here is an example of a product comparison table with a caption and summary to enhance accessibility and understanding:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Comparison Table</title>
</head>
<body>

    <h1 style="text-align: center;">Product Comparison</h1>

    <!-- Table with caption for title -->
    <table style="width: 80%; border: 1px solid black; border-collapse: collapse; margin: 20px auto;">
        <caption style="font-size: 1.5em; font-weight: bold; margin-bottom: 10px;">Smartphone
Comparison</caption>
        <thead>
            <tr>
                <th style="border: 1px solid black; padding: 8px; text-align: center;">Feature</th>
                <th style="border: 1px solid black; padding: 8px; text-align: center;">Product A</th>
                <th style="border: 1px solid black; padding: 8px; text-align: center;">Product B</th>
                <th style="border: 1px solid black; padding: 8px; text-align: center;">Product C</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td style="border: 1px solid black; padding: 8px; text-align: left;">Display</td>
                <td style="border: 1px solid black; padding: 8px; text-align: center;">6.1 inches</td>
```

```
            <td style="border: 1px solid black; padding: 8px; text-align: center;">6.4 inches</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">5.8 inches</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: left;">Camera</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">12 MP</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">48 MP</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">16 MP</td>
        </tr>
        <tr>
            <td style="border: 1px solid black; padding: 8px; text-align: left;">Battery</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">3000 mAh</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">4000 mAh</td>
            <td style="border: 1px solid black; padding: 8px; text-align: center;">3500 mAh</td>
        </tr>
    </tbody>
</table>

<!-- Adding summary for accessibility -->
<details>
    <summary style="font-size: 1.2em;">Click here for a summary of the products...</summary>
    <p>This table compares the features of three different smartphones: Product A, Product B, and Product C. The comparison includes display size, camera quality, and battery capacity, helping you make an informed choice when purchasing a new phone.</p>
</details>

</body>
</html>
```

Expected Output:

Product Comparison

Smartphone Comparison

Feature	Product A	Product B	Product C
Display	6.1 inches	6.4 inches	5.8 inches
Camera	12 MP	48 MP	16 MP
Battery	3000 mAh	4000 mAh	3500 mAh

Click here for a summary of the products...

Explanation of the Code:

1. **<caption>**:
    - The <caption> tag is used to provide a title to the table. In this case, "Smartphone Comparison" is used to describe the table's content.
    - The caption is styled with a larger font size (font-size: 1.5em), bold text (font-weight: bold), and some margin below it to visually separate it from the table.
2. **<summary>**:
    - The <summary> tag is used inside a <details> element to provide a summary of the table's content.
    - The summary is meant to be clicked by users to view more details about the table, especially for those using screen readers or assistive technology.
    - The <details> element is collapsible, so clicking the "Click here for a summary of the products" will reveal more information about the table.
3. **Table Structure**:
    - The table contains columns for "Feature", "Product A", "Product B", and "Product C".
    - Each row compares one feature of the products, like display size, camera quality, and battery capacity.

Benefits of Using <caption> and <summary>:

- **<caption>**: Provides a clear title to the table, making it more accessible and understandable to users.
- **<summary>**: Provides a textual summary or description that is helpful for accessibility purposes, especially for users with screen readers. It enhances the user experience by explaining the table's content in more detail.

Summary:

- **<caption>** is used to add a visible title to the table.
- **<summary>** inside a <details> element provides an accessible description that explains the table's content in more detail, improving usability for users with disabilities.